

The Ethical Concerns of Social Media Algorithms

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English 12

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February 4, 2026

Abstract

Meta's engagement-based social media algorithms greatly influence mental health, especially addiction, anxiety, and depression. These algorithms use machine learning to personalize a user's feed, optimizing the screentime spent on the platform. The data that machine learning algorithms use is mainly user activity, such as comments, shares, and watch time per video or post. With this, algorithms can ensure keeping users on social media by showing likeable content. These algorithms create feedback loops, encouraging prolonged screentime and compulsive scrolling. Over time, these repeated behaviors without limitations can affect the brain through stimulation, targeting key regions, such as the prefrontal cortex and amygdala. This weakens impulsive control, increases emotional sensitivity, and leads to poorer decision-making. Teens and young adults are prone to these effects of social media, resulting in anxiety, depressive symptoms, and emotional instability. Ethical concerns arise for the responsibility and prioritization of its creators, profiting from the well-being of the users.

The Ethical Concerns of Social Media Algorithms

How do Meta's engagement-based social media algorithms affect mental health through addiction, anxiety, and depression? In the digital age, most social media platforms owned by Meta are commonly used in social life through the way we communicate, view information, or perceive ourselves. Social media algorithms analyze the users on the platform based on metrics and interests. This personalizes the user's feed, attracting them to the social media platform through displaying likeable content. Meta's social media algorithm, an AI system designed to personalize users' feeds and increase engagement, has a negative impact on mental health, raising ethical concerns about the responsibility of its creators.

How do these algorithms personalize content?

Through social media algorithms, they have engagement features ensuring users stay on the platform. As a user interacts on the platform, whether through liking, commenting, or sharing a post, the algorithms take this into account. Meta's algorithm utilizes machine learning, gathering user data based on activity or interactions. (Kleinlugtebeld, 2024) As a result, the data based on activity influences what you see on social media. The algorithm does this to receive a reaction out of the user, as they are most likely to interact with a post they see, relating to the user's hobbies or interests. This type of content is prioritized by the algorithm, ensuring to generate an emotional reaction for the user. Viewing likeable content can stir up a certain reaction, whether it makes you laugh or be surprised, which is the algorithm's main goal of personalizing your feed. (Arvind, 2023)

Furthermore, Algorithms operate based on predictions while utilizing machine learning. Algorithms collect data on user activity to determine what they want to see next. (Pellegrino et al., 2022). This is known as predictive modeling, where the algorithm analyzes patterns based on

user activity. Predictive modeling within social media algorithms ensures that feeds aren't random, making platforms more relevant to the user.

Social media's cognitive impact

With social media algorithms displaying likeable content on your feed, it can lead to addictive use, rewiring our brains differently. A feedback loop is induced through social media by the constant promise of viewing rewardable, personalized content. (De et al., 2025) This sensation of reward is dopamine, where transmitters inform the brain to encourage certain behaviors. Specifically, dopamine enforces the feeling of pleasure and motivation for the user. The feeds that algorithms personalize for us stimulate our brains to release dopamine through viewing content that relates to us. (Author, 2024) Through feed personalization, social media algorithms can exploit our dopamine reward system.

Furthermore, excessive stimulation of dopamine can affect how the brain deals with rewards. This is because repeated actions, such as constant dopamine, can turn everyday tasks to feel less exciting. (Amiri, 2025) The brain craves social media over normal tasks. Specifically, repeated engagement in social media affects our prefrontal cortex in terms of decision-making. Since rewards are unpredictable through dopamine, users continuously scroll, leading to unhealthy amounts of screentime.

Unhealthy increase in screen time

Repeated dopamine stimulation can lead to an increase in screentime. Meta platforms, like Instagram or Facebook, encourage endless scrolling through reels. Every video on Reels is short, thirty seconds to a minute long. Endless scrolling is unlimited on these platforms, which can shorten attention spans and shift focus away from important tasks. (De et al., 2025) This correlates with dopamine stimulation; since every video is short, our brain is rewarded with

something new each scroll. On top of this, the algorithm personalizes the user's feed based on their likes, substituting dopamine-reward circuits in support of addictive use. (Qiu, 2021)

Although endless scrolling seems harmless, we are constantly stimulated by dopamine, which can influence how we regulate our emotions. Excessive engagement on social media can contribute to anxiety, insomnia, or even depression

Affect on mental health

Moreover, addiction to social media results to increase in emotional sensitivity. They can impact our decision-making, inhibitory control, and most importantly, how we control our emotions. Research proves that cutting down on social media can reduce anxiety by about 16%, depression by about 25%, and insomnia by about 15% for young adults. (Berman, 2025) These results prove the impact that social media has on our mental health. Additionally, the influence of the algorithm can contribute to several factors affecting mental health.

Since social media algorithms constantly personalize our feeds, they align with the idea of doomscrolling, as we're constantly rewarded with dopamine. This factor contributes to our mental health, as our inhibitory control weakens, which can affect those facing anxiety or depression. The prolonged effects of dopamine pathways on our prefrontal cortex and amygdala lead to poorer cognitive decision-making and a weakened suppression of certain impulsive behaviors. (De et al., 2025) These results are prone to increasing our emotional sensitivity to react or view differently.

During the COVID-19 pandemic, more users significantly became active on Meta platforms, as a form of entertainment and support. Social media can be viewed as a coping mechanism for receiving emotional support, experiencing poor psychological adjustment due to overload from feeds generated by the algorithm. (Tunca et al., 2025) This shows the ineffectivity

relying on social media for emotional support, as users can become overwhelmed viewing too much information or emotionally intense content. Overall, these findings show that Meta's engagement system not only fuels compulsive behavior but also contributes to the decline of emotional regulation.

Conclusion

Meta's social media algorithm can impact mental health in various ways, such as how we experience dopamine, maximize screen time usage, and control our emotions. The consequences of personalization lead to these factors that users experience. Algorithm personalization focuses on the data of the user based on activity, as its primary goal is to predict what content is likely to receive a user reaction. Cognitively, dopamine reward is repeatedly stimulated through feeds designed to be likable, affecting our prefrontal cortex. Through dopamine, endless scrolling becomes a habit, turning behavioral patterns into unhealthy addiction. Lastly, personalization has affects to our mental health in ways, such as utilizing social media as a coping mechanism. Although many adolescents are still undergoing major development with their frontal cortex and amygdala, affecting the way they think or react, additional studies are needed to show how these specific neurological aspects correlate with addictive social media use. The majority of these studies show correlations between social media and cognitive functioning, but lack information regarding changes in the structure of the brain. More research is needed to determine whether excessive dopamine stimulation impacts reward-processing circuits that can alter the gray matter density during the adolescent development. Social media algorithms designed by Meta prioritize user engagement on platforms instead of their users' well-being, raising ethical concerns about its creator's responsibility.

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